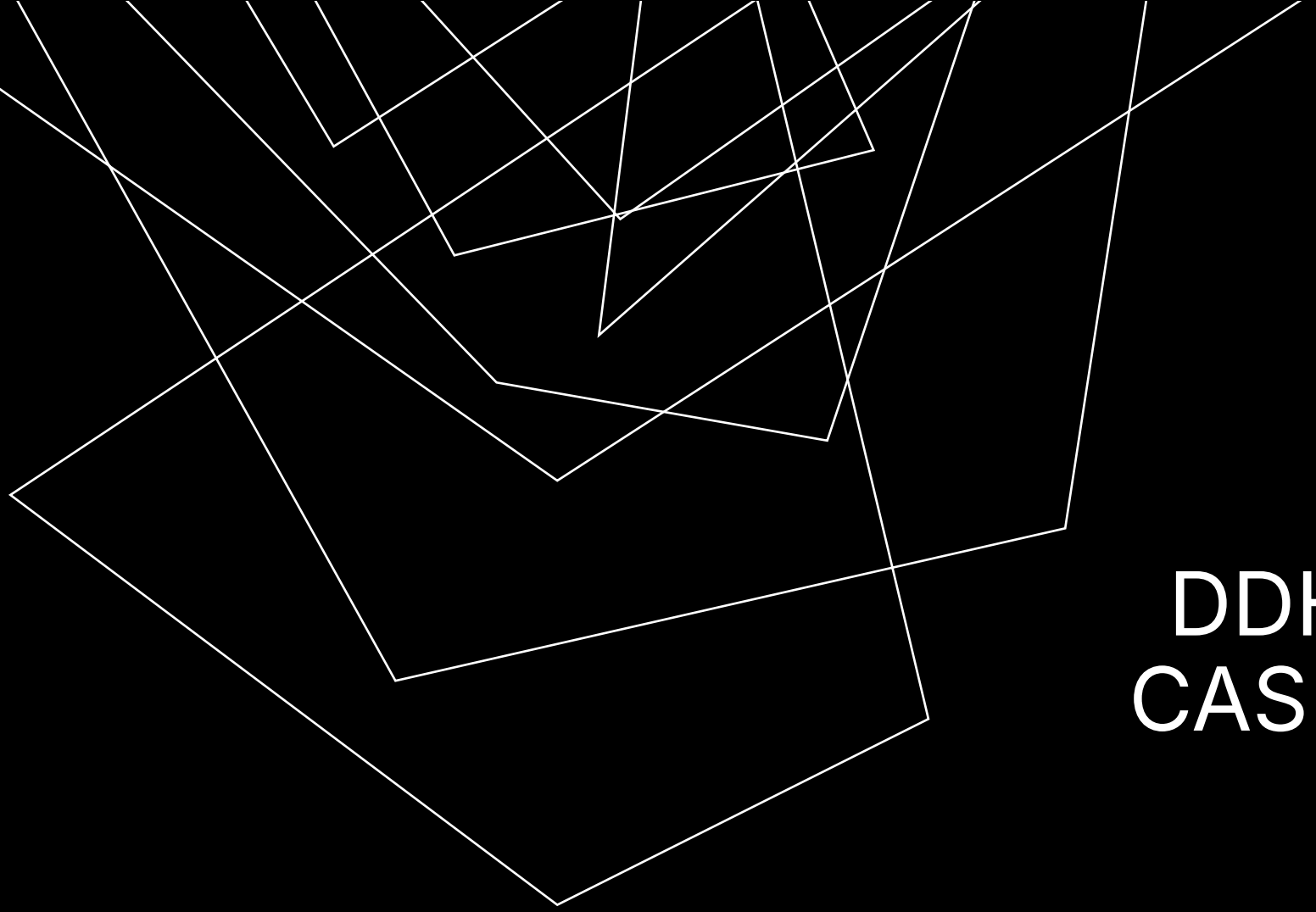


DDH NETWORKS CASE STUDY 2025

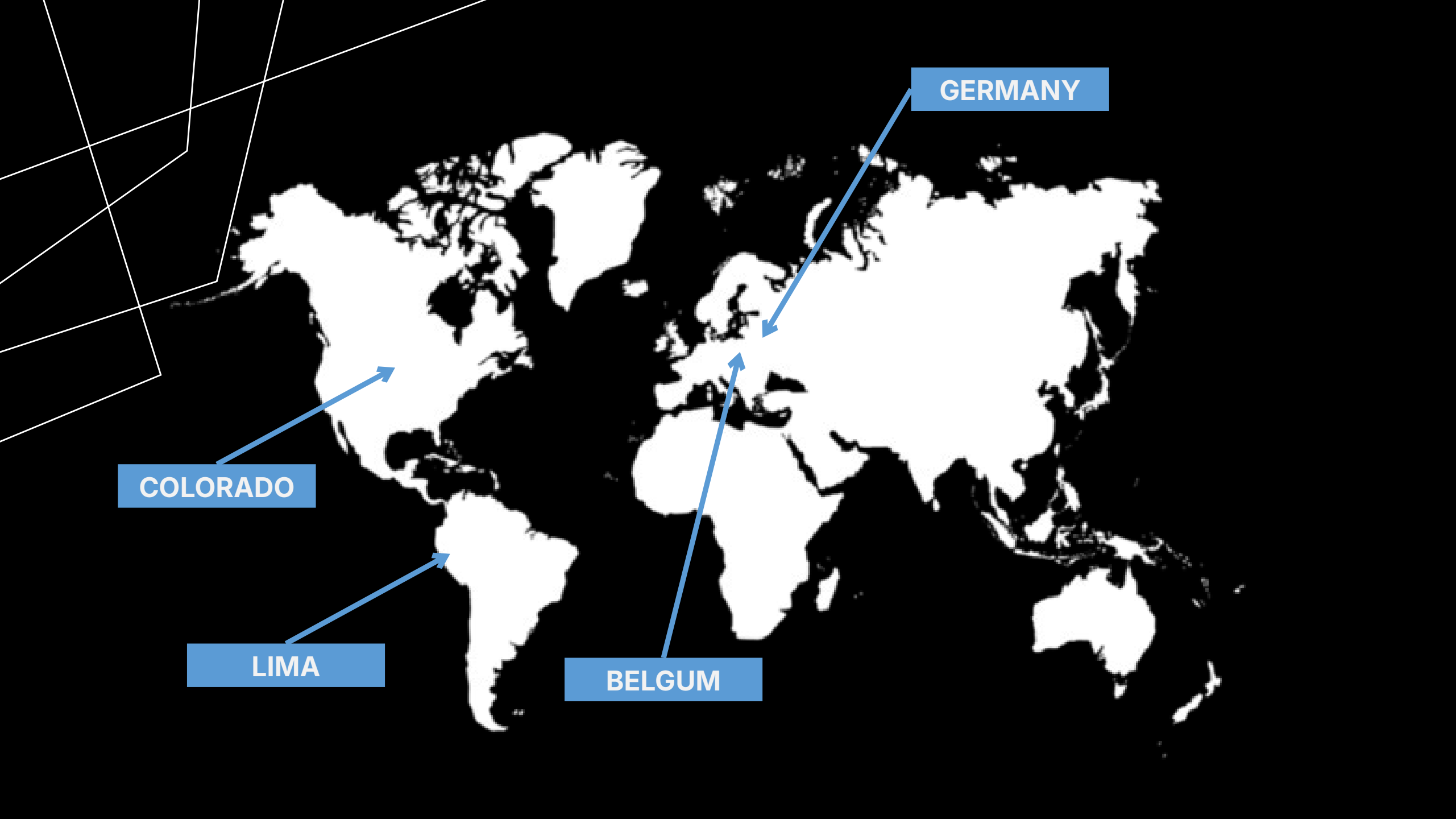
Presentation by
Darren D
Huynh

OBJECTIVES

- Discuss VLSM Scheme and Logical Topology
- Configure OSPFv2 and RIP for routing
- Configure GRE and Secure GRE Tunnel with IPsec
- Discuss NAT and Germany ACL
- DHCP and VLAN Configurations
- Verification and Testing
- Questions



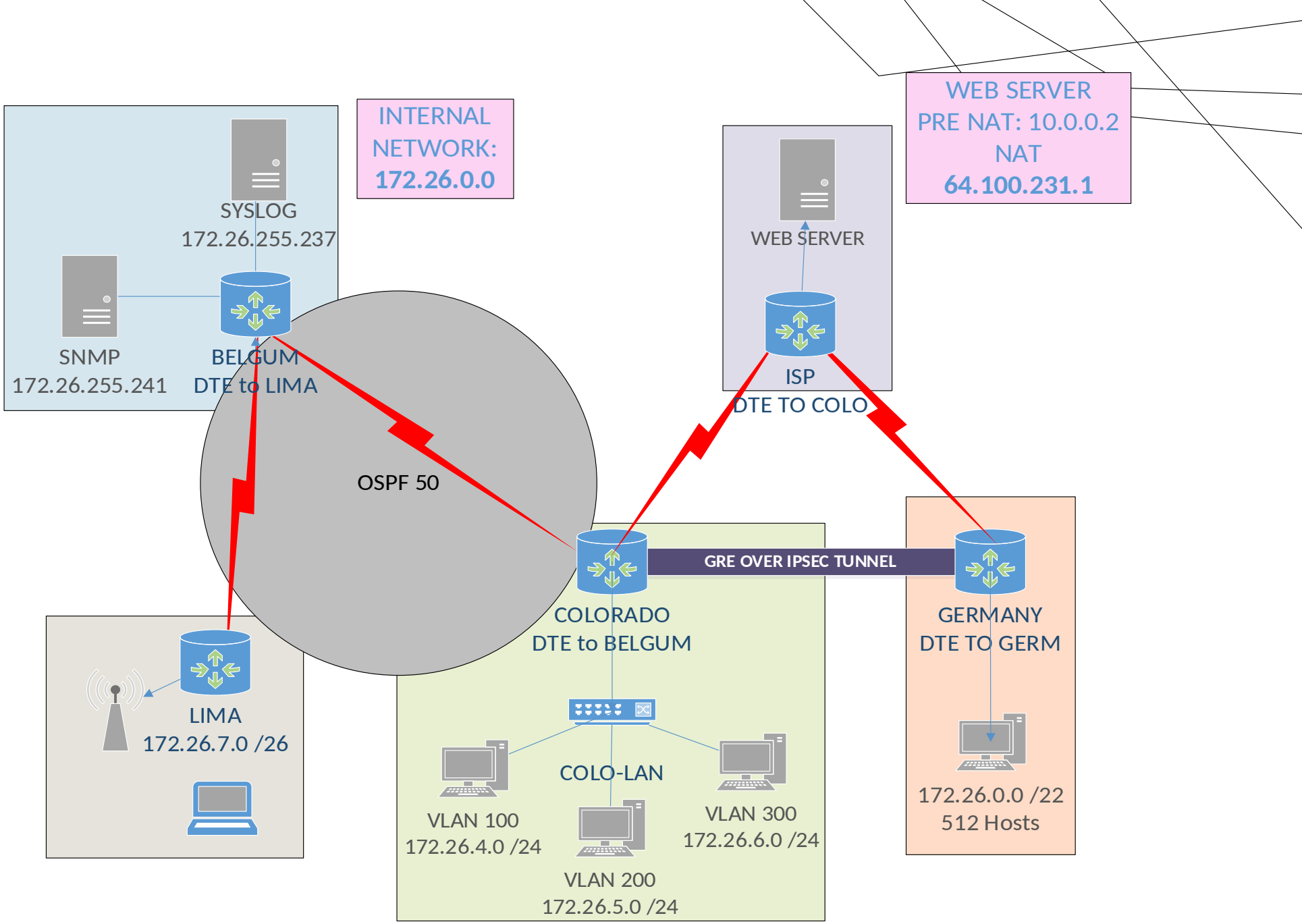
PHASE 1: LAN/WAN AND INTERFACES


GERMANY
COLORADO
LIMA
BELGUM

VLSM SCHEME

IP ADDRESS	SUBNET	CIDR	ACTIVE	NAME	REQUIRED HOST
172.26.0.0	255.255.252.0	/22	YES	GERMANY	512
172.26.4.0	255.255.255.0	/24	YES	COLORADO VLAN 100	250
172.26.5.0	255.255.255.0	/24	YES	COLORADO VLAN 200	250
172.26.6.0	255.255.255.0	/24	YES	COLORADO VLAN 300	250
172.26.7.0	255.255.255.192	/26	YES	LIMA	50
172.26.255.236	255.255.255.252	/30	YES	SYSLOG SERVER	2
172.26.255.240	255.255.255.252	/30	YES	SNMP SERVER	2
172.26.255.244	255.255.255.252	/30	YES	MANAGEMENT	2
	255.255.255.2			LINK to	

LOGICAL TOPOLOGY



COLORADO INTERFACES

ROUTER	INTERFACES	IP ADDRESS	CIDR	CONNECTION TYPE	ACTIVE	DESCRIPTION
COLORADO	G0/0/1	N/A	N/A	ETHERNET	YES	LINK to COLO-LAN
COLORADO	G0/0/1.100	172.26.4.1	/24	ENCAPSULATED	YES	LINK to VLAN 100
COLORADO	G0/0/1.200	172.26.5.1	/24	ENCAPSULATED	YES	LINK to VLAN 200
COLORADO	G0/0/1.300	172.26.6.1	/24	ENCAPSULATED	YES	LINK to VLAN 300
COLORADO	G0/0/1.401	172.26.255.245	/30	ENCAPSULATED	YES	LINK to VLAN 401
COLORADO	S0/1/0	209.165.200.1	/30	SERIAL	YES	LINK to ISP
COLORADO	S0/1/1	172.26.255.250	/30	SERIAL	YES	LINK to BELGUM
COLORADO	Tunnel0	1.1.1.1	/30	TUNNEL	YES	TUNNEL to GERMANY
COLORADO	Lo0	10.10.10.10	/32	LOOPBACK	YES	LOOPBACK for COLORADO

GERMANY INTERFACES

ROUTER	INTERFACES	IP ADDRESS	CIDR	CONNECTION TYPE	ACTIVE	DESCRIPTION
GERMANY	G0/0/0	172.26.0.0	/22	ETHERNET	YES	LINK to GERMANY-LAN
GERMANY	S0/1/0	209.165.100.2	/30	SERIAL	YES	LINK to ISP
GERMANY	Tunnel0	1.1.1.2	/30	TUNNEL/SERIAL	YES	TUNNEL to COLORADO

LIMA INTERFACES

ROUTER	INTERFACES	IP ADDRESS	CIDR	CONNECTION TYPE	ACTIVE	DESCRIPTION
LIMA	G0/0/0	172.26.7.1	/26	ETHERNET	YES	LINK to LIMA-LAN
LIMA	S0/1/0	172.26.255.254	/30	SERIAL	YES	LINK to BELGUM
LIMA	Lo0	30.30.30.30	/32	LOOPBACK	YES	LOOPBACK for LIMA

BELGUM INTERFACES

ROUTER	INTERFACES	IP ADDRESS	CIDR NOTATION	CONNECTION TYPE	ACTIVE	DESCRIPTION
BELGUM	G0/0/0	172.26.255.237	/30	ETHERNET	YES	LINK to SYSLOG
BELGUM	G0/0/1	172.26.255.241	/30	ETHERNET	YES	LINK to SNMP
BELGUM	S0/1/0	172.26.255.249	/30	SERIAL	YES	LINK to COLORADO
BELGUM	S0/1/1	172.26.255.253	/30	SERIAL	YES	LINK to LIMA
BELGUM	Lo0	20.20.20.20	/32	LOOPBACK	YES	LOOPBACK for BELGUM

PHASE 2: OSPFV2 AND RIPV2

OSPFV2 AND RIPV2

```
router ospf 50
 router-id 10.10.10.10
 redistribute rip metric 1
 network 172.26.4.0 0.0.0.255 area 0
 network 172.26.5.0 0.0.0.255 area 0
 network 172.26.6.0 0.0.0.255 area 0
 network 172.26.255.244 0.0.0.3 area 0
 network 172.26.255.248 0.0.0.3 area 0
 default-information originate
```

COLORADO

GERMANY

```
router rip
 version 2
 network 1.0.0.0
 network 172.26.0.0
 no auto-summary
```

```
router rip
 version 2
 redistribute ospf 50 metric 1
 network 1.0.0.0
 network 172.26.0.0
 no auto-summary
```

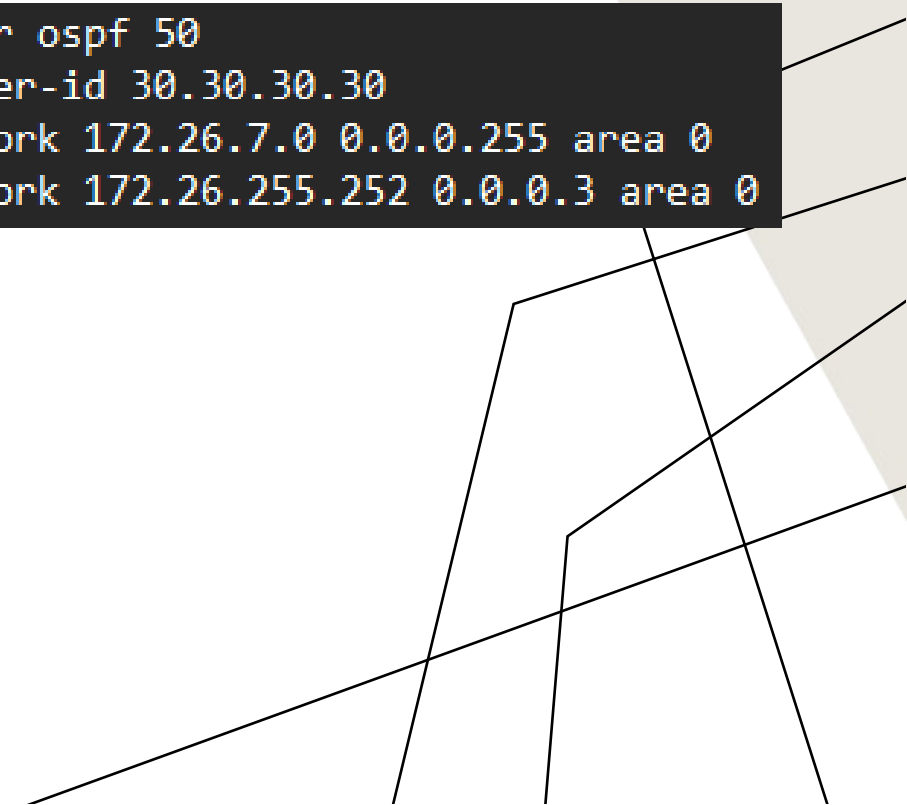
OSPFV2 AND RIPV2

```
router ospf 50
 router-id 20.20.20.20
 network 172.26.255.236 0.0.0.3 area 0
 network 172.26.255.240 0.0.0.3 area 0
 network 172.26.255.248 0.0.0.3 area 0
 network 172.26.255.252 0.0.0.3 area 0
```

BELGUM

LIMA

```
router ospf 50
 router-id 30.30.30.30
 network 172.26.7.0 0.0.0.255 area 0
 network 172.26.255.252 0.0.0.3 area 0
```



PHASE 3: CONFIGURATIONS

NAT WITH PAT CONFIGS

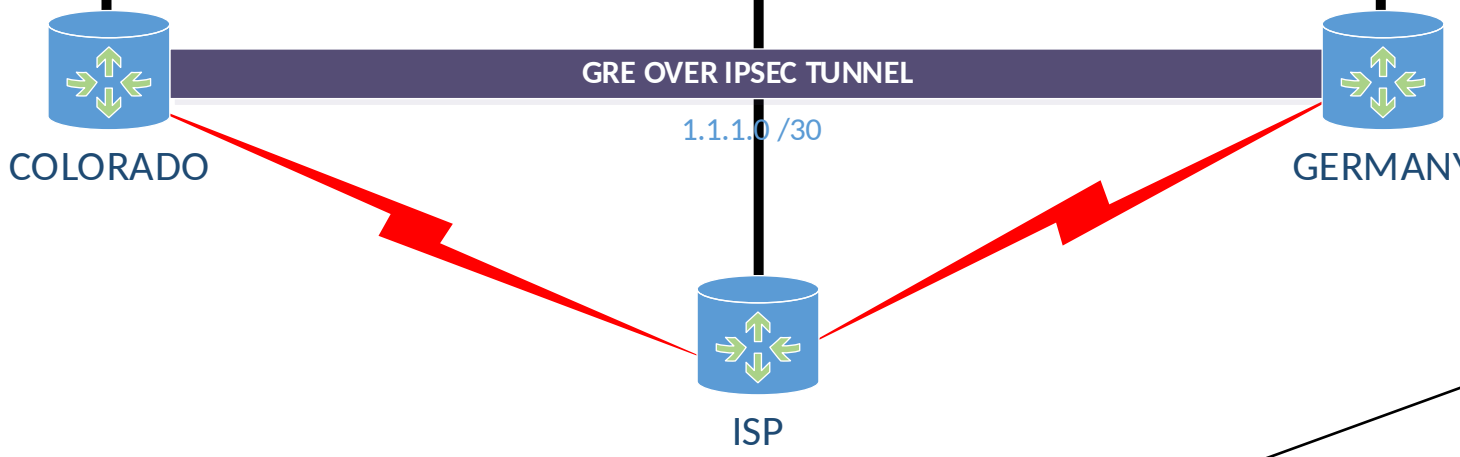
```
ip nat pool COLO-NAT 55.69.241.1 55.69.241.7 netmask 255.255.255.248
ip nat inside source list 10 pool COLO-NAT overload
```

```
ip access-list standard 10
 10 permit 172.26.0.0 0.0.255.255
```

```
ip nat pool GERMANY-NAT 77.100.10.1 77.100.10.1 netmask 255.255.255.255
ip nat inside source list 10 pool GERMANY-NAT overload
```

```
ip access-list standard 10
 10 permit 172.26.0.0 0.0.255.255
```

```
ip nat inside source static 10.0.0.2 64.100.231.1
```



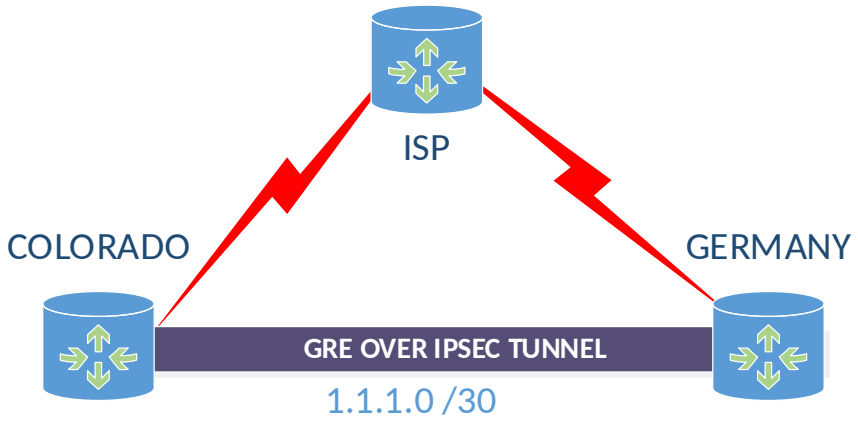
NAT WITH PAT CONFIGS

```
interface Serial0/1/0
 no shutdown
 description LINK to ISP
 ip address 209.165.200.1 255.255.255.252
 ip nat outside
 crypto map GREMAP
```

```
interface Serial0/1/1
 no shutdown
 description LINK TO BELGUM
 ip address 172.26.255.250 255.255.255.252
 ip nat inside
```

```
interface GigabitEthernet0/0/1
 no shutdown
 no ip address
 ip nat inside
 negotiation auto
```

```
interface GigabitEthernet0/0/1.100
 description VLAN 100
 encapsulation dot1Q 100
 ip address 172.26.4.1 255.255.255.0
 ip nat inside
```

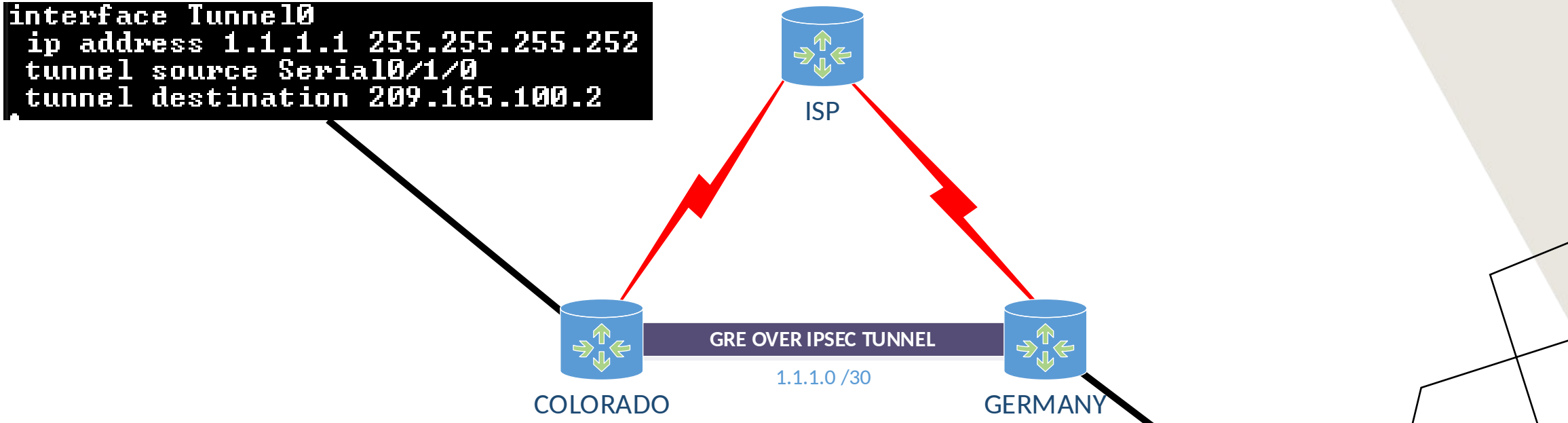



```
interface Serial0/1/0
 no shutdown
 description LINK to ISP
 ip address 209.165.100.2 255.255.255.252
 ip nat outside
 crypto map GREMAP
```

```
interface GigabitEthernet0/0/0
 no shutdown
 ip address 172.26.0.1 255.255.252.0
 ip nat inside
 ip access-group GERMANY-ACL in
 negotiation auto
```

GRE TUNNEL

```
interface Tunnel0
 ip address 1.1.1.1 255.255.255.252
 tunnel source Serial0/1/0
 tunnel destination 209.165.100.2
```




```
interface Tunnel0
 ip address 1.1.1.2 255.255.255.252
 tunnel source Serial0/1/0
 tunnel destination 209.165.200.1
```

IPSEC CONFIGS

ISAKMP POLICY

```
crypto isakmp policy 10
 encryption aes 256
 hash sha256
 authentication pre-share
 group 19
 lifetime 3600
```

TRANSFORM SET

```
crypto ipsec transform-set TRANSET esp-aes 256 esp-sha256-hmac
 mode tunnel
```

COLORADO KEY

```
crypto isakmp key GREKEY address 209.165.100.2
```

GERMANY KEY

```
crypto isakmp key GREKEY address 209.165.200.1
```

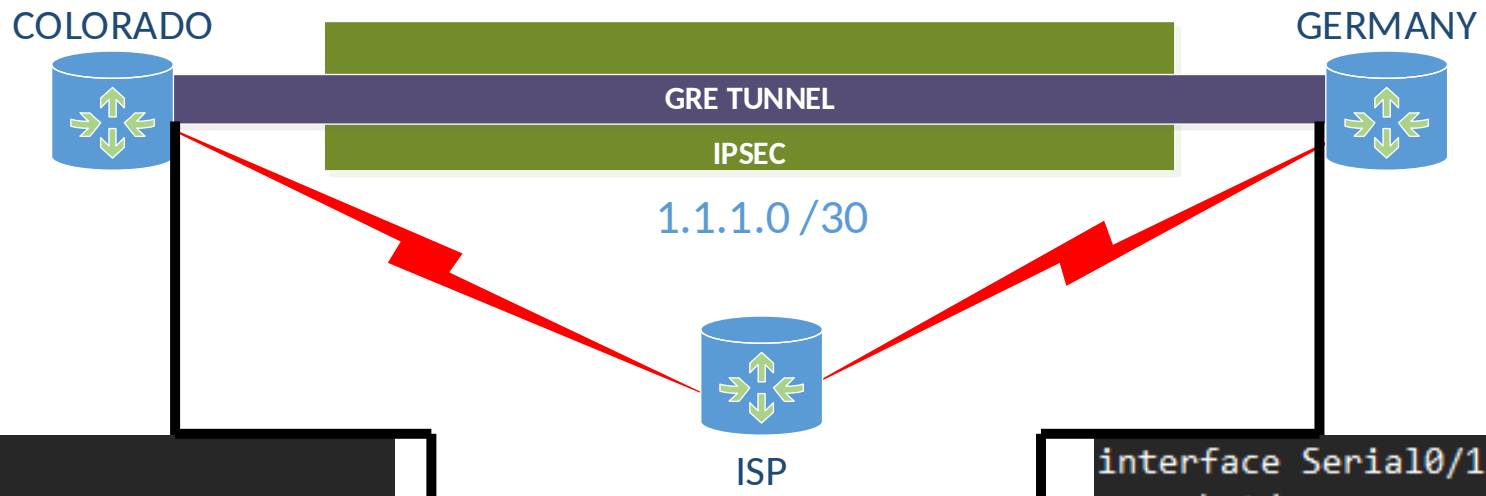
COLO CRYPTO MAP

```
crypto map GREMAP 10 ipsec-isakmp
 set pfs group19
 set peer 209.165.100.2
 set transform-set TRANSET
 match address GRE-VPN
```

GERMANY CRYPTO MAP

```
crypto map GREMAP 10 ipsec-isakmp
 set pfs group19
 set peer 209.165.200.1
 set transform-set TRANSET
 match address GRE-VPN
```

IPSEC CONFIGS




```
interface Serial0/1/0
 no shutdown
 description LINK to ISP
 ip address 209.165.200.1 255.255.255.252
 ip nat outside
 crypto map GREMAP
```

```
ip access-list extended GRE-VPN
 remark ACL FOR TRAFFIC TAGGING
 permit ip 172.26.0.0 0.0.255.255 172.26.0.0 0.0.3.255
```

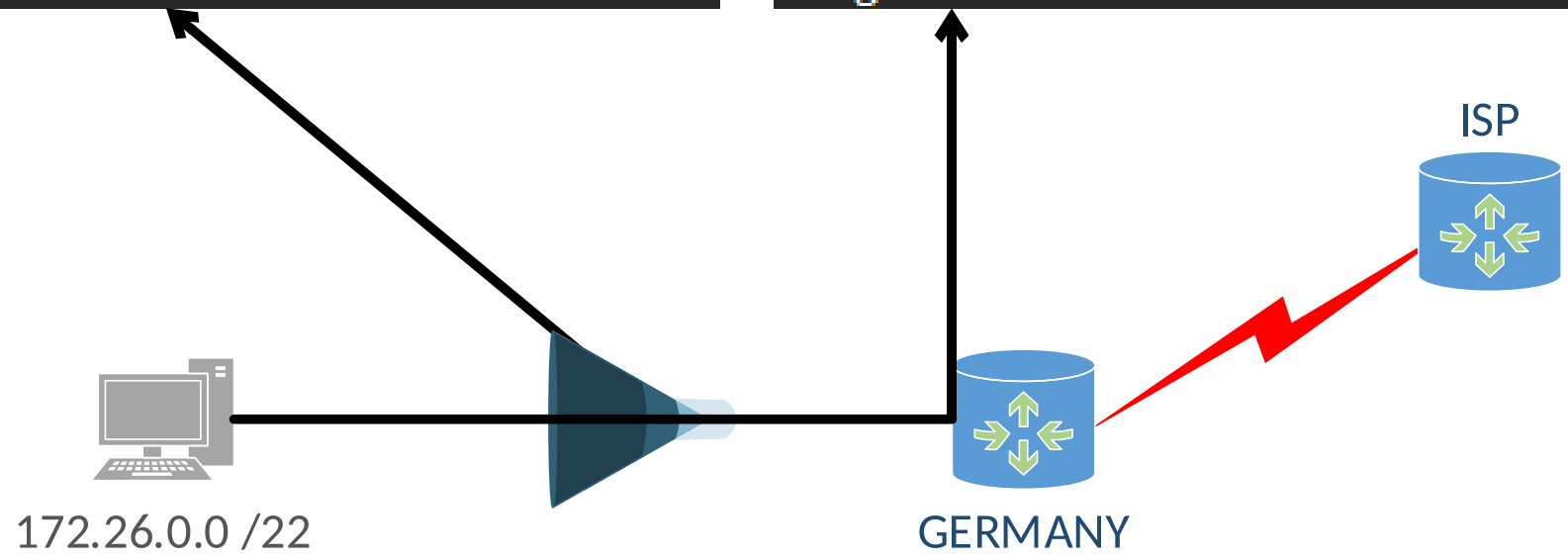
```
interface Serial0/1/0
 no shutdown
 description LINK to ISP
 ip address 209.165.100.2 255.255.255.252
 ip nat outside
 crypto map GREMAP
```

```
ip access-list extended GRE-VPN
 remark ACL FOR TRAFFIC TAGGING
 permit ip 172.26.0.0 0.0.3.255 172.26.0.0 0.0.255.255
```

GERMANY ACL

```
ip access-list extended GERMANY-ACL
 10 remark CONTROLS LAN TRAFFIC
 10 permit tcp 172.26.0.0 0.0.3.255 host 209.165.200.1 eq 22
 20 permit ip 172.26.0.0 0.0.3.255 172.26.0.0 0.0.255.255
 30 permit tcp 172.26.0.0 0.0.3.255 host 64.100.231.1 eq www
 40 permit udp any eq bootpc any
 50 permit udp any eq bootps any
 60 deny ip any any
```

```
interface GigabitEthernet0/0/0
 no shutdown
 ip address 172.26.0.1 255.255.252.0
 ip nat inside
 ip access-group GERMANY-ACL in
 negotiation auto
```



PHASE 3: DHCP AND VLAN

DHCP POOLS

POOL	NETWORK	SUBNET	EXCLUDED-ADDRESS	ADDRESS RANGE	DOMAIN NAME	DEFAULT ROUTER
COLO VLAN 100	172.26.4.0	/24	172.26.4.1-.10	172.26.4.11-254	COLO.com	172.26.4.1
COLO VLAN 200	172.26.5.0	/24	172.26.5.1-.10	172.26.5.11-.254	COLO.com	172.26.5.1
COLO VLAN 300	172.26.6.0	/24	172.26.6.1-.10	172.26.6.11-.254	COLO.com	172.26.6.1
GERMANY	172.26.0.0	/22	172.26.0.1-.20	172.26.0.21 - 172.26.3.254	GERM.com	172.26.0.1
LIMA	172.26.7.0	/26	172.26.7.1-.10	172.26.7.11-.254	LIMA.com	172.26.7.1

DHCP CONFIGS

```
ip dhcp excluded-address 172.26.4.1 172.26.4.10
ip dhcp excluded-address 172.26.5.1 172.26.5.10
ip dhcp excluded-address 172.26.6.1 172.26.6.10
!
ip dhcp pool COLO-VLAN100
 network 172.26.4.0 255.255.255.0
 default-router 172.26.4.1
 domain-name COLO.com
!
ip dhcp pool COLO-VLAN200
 network 172.26.5.0 255.255.255.0
 default-router 172.26.5.1
 domain-name COLO.com
!
ip dhcp pool COLO-VLAN300
 network 172.26.6.0 255.255.255.0
 default-router 172.26.6.1
 domain-name COLO.com
```

```
ip dhcp excluded-address 172.26.0.1 172.26.0.20
!
ip dhcp pool GERMANY-DHCP
 network 172.26.0.0 255.255.252.0
 default-router 172.26.0.1
```

```
ip dhcp excluded-address 172.26.7.1 172.26.7.10
!
ip dhcp pool LIMA
 network 172.26.7.0 255.255.255.192
 default-router 172.26.7.1
```

VLANS

```
interface GigabitEthernet1/0/1
 switchport access vlan 100
 switchport mode access
 switchport port-security maximum 3
 switchport port-security mac-address sticky
 switchport port-security
```

```
interface GigabitEthernet1/0/5
 switchport access vlan 200
 switchport mode access
 switchport port-security maximum 3
 switchport port-security mac-address sticky
 switchport port-security
```

```
interface GigabitEthernet1/0/9
 switchport access vlan 300
 switchport mode access
 switchport port-security maximum 3
 switchport port-security mac-address sticky
 switchport port-security
```

```
interface GigabitEthernet1/0/13
 switchport access vlan 500
 switchport mode access
 switchport port-security maximum 3
 switchport port-security mac-address sticky
 switchport port-security
 shutdown
```

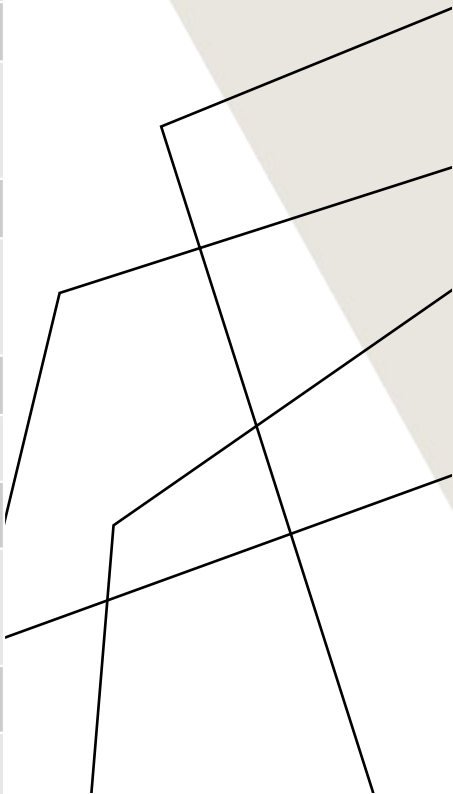
```
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active
100  VLAN0100                         active    Gi1/0/1, Gi1/0/2, Gi1/0/3
                                                Gi1/0/4
200  VLAN0200                         active    Gi1/0/5, Gi1/0/6, Gi1/0/7
                                                Gi1/0/8
300  VLAN0300                         active    Gi1/0/9, Gi1/0/10, Gi1/0/11
                                                Gi1/0/12
401  Management                       active
500  Native                           active    Gi1/0/13, Gi1/0/14, Gi1/0/15
                                                Gi1/0/16, Gi1/0/17, Gi1/0/18
                                                Gi1/0/19, Gi1/0/20, Gi1/0/21
                                                Gi1/0/22, Gi1/0/23, Gi1/0/25
                                                Gi1/0/26, Gi1/0/27, Gi1/0/28
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
```

MANAGEMENT VLAN:
192.168.255.246

PHASE 4: TESTING AND VERIFICATION

TESTING AND VERIFICATIO N

SOURCE	DESTINATION	PROTOCOL	EXPECTED RESULT
Host on VLAN1	ISP	HTTP	PASS
Host on VLAN1	Host on Germany LAN	ICMP	PASS
Host on VLAN1	Host on VLAN 2	ICMP	PASS
Host on VLAN1	Host on VLAN 3	ICMP	PASS
Host on VLAN 2	Host on VLAN 3	ICMP	PASS
Host on VLAN 2	Host on Germany LAN	ICMP	PASS
Host on VLAN 2	ISP	HTTP	PASS
Host on VLAN 3	Host on Germany LAN	ICMP	PASS
Host on VLAN 3	ISP	HTTP	PASS
Host on Germany	ISP	HTTP	PASS
Host on Germany	ISP	Telnet	FAIL
Router on Colorado	SYSLog Server	LOGS	PASS
NAT on Germany		NAT	PASS
NAT on Colorado		NAT	PASS



NAT VERIFICAITON



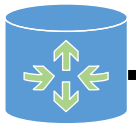
ISP

```
ISP#sh ip nat translations
Pro  Inside global         Inside local       Outside local      Outside global
---  64.100.231.1          10.0.0.2           ---                ---
icmp 64.100.231.1:1        10.0.0.2:1         55.69.241.1:1      55.69.241.1:1
Total number of translations: 2
```



GERMANY

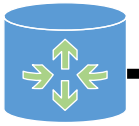
```
GERMANY#sh ip nat translations
Pro  Inside global         Inside local       Outside local      Outside global
tcp  77.100.10.1:1064      172.26.0.22:1064   64.100.231.1:80    64.100.231.1:80
tcp  77.100.10.1:1063      172.26.0.22:1063   64.100.231.1:80    64.100.231.1:80
tcp  77.100.10.1:1065      172.26.0.22:1065   64.100.231.1:80    64.100.231.1:80
Total number of translations: 3
```



COLORADO

```
COLORADO#sh ip nat translations
Pro  Inside global         Inside local       Outside local      Outside global
icmp 55.69.241.1:1         172.26.4.12:1      64.100.231.1:1     64.100.231.1:1
Total number of translations: 1
```

IPSEC VERIFICATION



COLORADO

```
COLORADO#sh crypto ipsec sa

interface: Serial0/1/0
    Crypto map tag: GREMAP, local addr 209.165.200.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
   remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
   current_peer 209.165.100.2 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 18, #pkts encrypt: 18, #pkts digest: 18
    #pkts decaps: 17, #pkts decrypt: 17, #pkts verify: 17
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 209.165.200.1, remote crypto endpt.: 209.165.100.2
     plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
     current outbound spi: 0xE23AA18A(3795493258)
     PFS (Y/N): Y, DH group: group19

     inbound esp sas:
      spi: 0x9B804ED(163054829)
        transform: esp-256-aes esp-sha256-hmac ,
        in use settings ={Tunnel, }
        conn id: 2001, flow_id: ESG:1, sibling_flags FFFFFFFF80004048, crypto map: GREMAP, initiator : True
         sa timing: remaining key lifetime (k/sec): (4607997/3506)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE(ACTIVE)

COLORADO#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
209.165.100.2   209.165.200.1   QM_IDLE           1001 ACTIVE
```

IPSEC VERIFICATION



GERMANY

```
GERMANY#sh crypto ipsec sa

interface: Serial0/1/0
    Crypto map tag: GREMAP, local addr 209.165.100.2

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
   remote ident (addr/mask/prot/port): (0.0.0.0/0.0.0.0/47/0)
   current_peer 209.165.200.1 port 500
     PERMIT, flags={origin_is_acl,}
    #pkts encaps: 25, #pkts encrypt: 25, #pkts digest: 25
    #pkts decaps: 25, #pkts decrypt: 25, #pkts verify: 25
    #pkts compressed: 0, #pkts decompressed: 0
    #pkts not compressed: 0, #pkts compr. failed: 0
    #pkts not decompressed: 0, #pkts decompress failed: 0
    #send errors 0, #recv errors 0

     local crypto endpt.: 209.165.100.2, remote crypto endpt.: 209.165.200.1
     plaintext mtu 1438, path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
     current outbound spi: 0x9B804ED(163054829)
     PFS (Y/N): Y, DH group: group19

     inbound esp sas:
      spi: 0xE23AA18A(3795493258)
        transform: esp-256-aes esp-sha256-hmac ,
        in use settings ={Tunnel, }
        conn id: 2001, flow_id: ESG:1, sibling_flags FFFFFFFF80000048, crypto map: GREMAP, initiator : False
         sa timing: remaining key lifetime (k/sec): (4607996/3304)
        IV size: 16 bytes
        replay detection support: Y
        Status: ACTIVE(ACTIVE)

GERMANY#sh crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id status
209.165.100.2   209.165.200.1   QM_IDLE           1001 ACTIVE

IPv6 Crypto ISAKMP SA
```

GERMANY VERIFICATION

SOURCE	DESTINATION	PROTOCOL	RESULT
Host on Germany	ISP	ICMP	FAIL
Host on Germany	ISP	HTTP	PASS
Host on Germany	LIMA	ICMP	PASS
Host on Germany	SNMP	ICMP	PASS
Host on Germany	SYSLOG	ICMP	PASS
Host on Germany	VLAN 100	ICMP	PASS
Host on Germany	VLAN 200	ICMP	PASS

GERMANY VERIFICATION

```
Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2293:f41:798:c43a%15
   IPv4 Address. . . . . . . . . . . : 172.26.0.22
   Subnet Mask . . . . . . . . . . . : 255.255.252.0
   Default Gateway . . . . . . . . . : fe80::3%15
                                       172.26.0.1

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Ethernet adapter Bluetooth Network Connection:

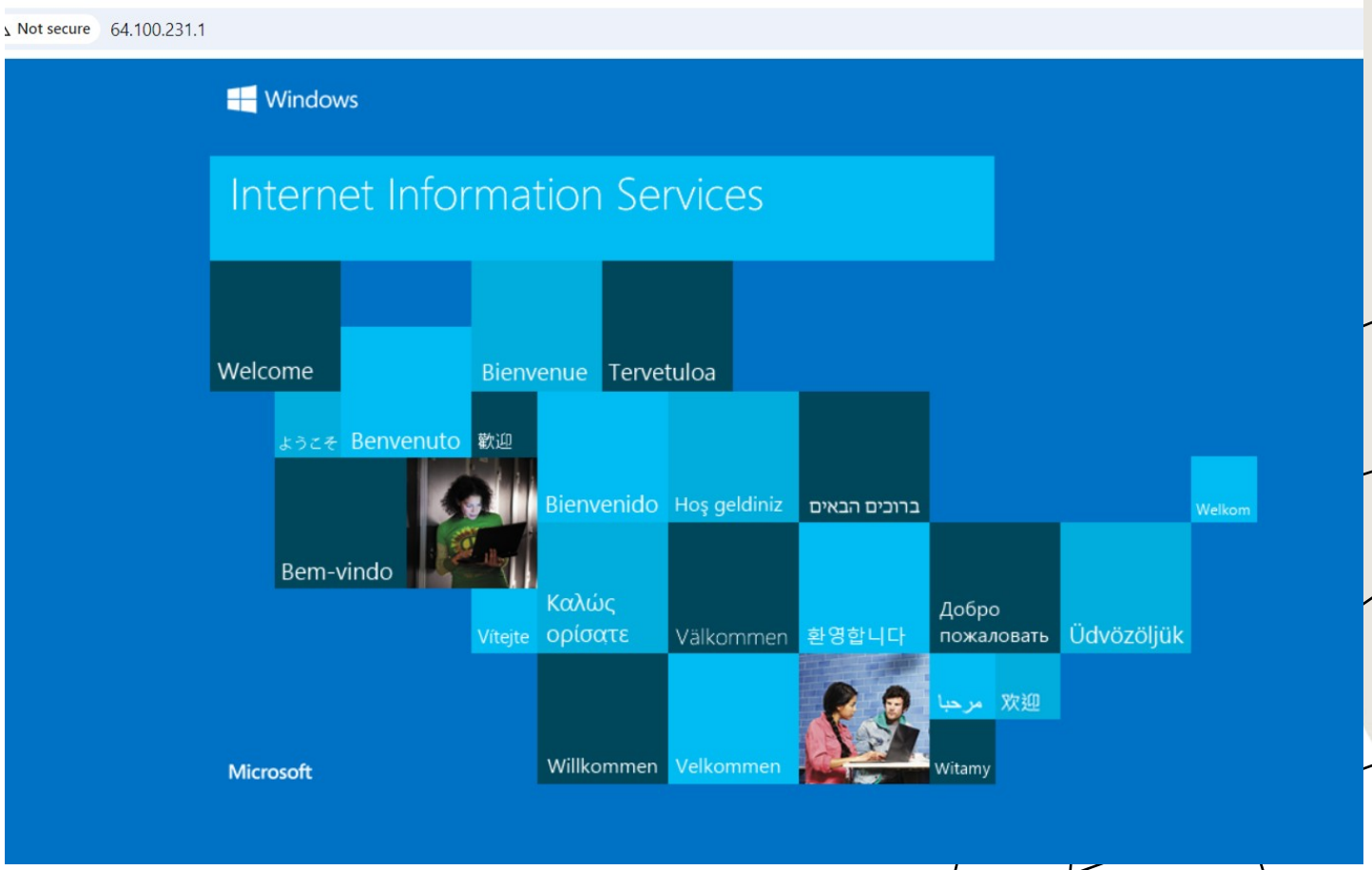
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : deltacampus.net

C:\Users\MH401-Student>ping 64.100.231.1

Pinging 64.100.231.1 with 32 bytes of data:
Reply from 172.26.0.1: Destination net unreachable.
Reply from 172.26.0.1: Destination net unreachable.
Reply from 172.26.0.1: Destination net unreachable.
Reply from 172.26.0.1: Destination net unreachable.
```



GERMANY VERIFICATIO N

```
C:\Users\Cisco Lab>telnet 209.165.100.1
Connecting To 209.165.100.1...Could not open connection to the host, on port 23:
 Connect failed
```

COLORADO VLAN 100 VERIFICATION

SOURCE	DESTINATION	PROTOCOL	EXPECTED RESULT
VLAN 100	ISP	HTTP	PASS
VLAN 100	LIMA	ICMP	PASS
VLAN 100	SNMP	ICMP	PASS
VLAN 100	SYSLOG	ICMP	PASS
VLAN 100	GERMANY	ICMP	PASS
VLAN 100	VLAN 200	ICMP	PASS
VLAN 100	VLAN 300	ICMP	PASS

COLORADO VLAN 100 VERIFICATION

```
Windows IP Configuration


Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . : COLO.com
   Link-local IPv6 Address . . . . . : fe80::658b:56dd:a53b:d5f3%11
   IPv4 Address. . . . . . . . . . . : 172.26.4.12
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 172.26.4.1

Tunnel adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

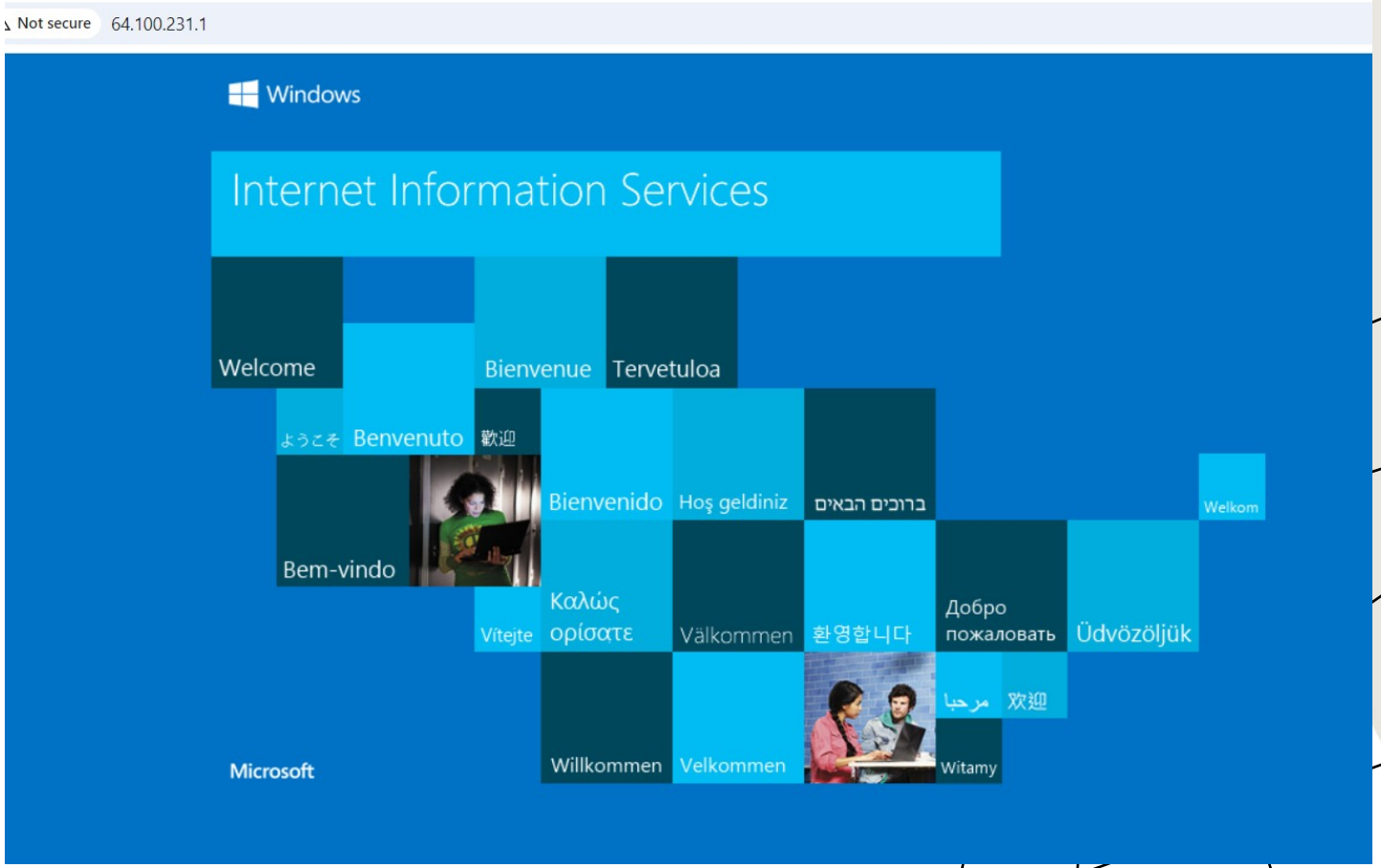
Tunnel adapter isatap.COLO.com:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . : COLO.com

C:\Users\Cisco Lab>ping 172.26.0.22

Pinging 172.26.0.22 with 32 bytes of data:
Reply from 172.26.0.22: bytes=32 time=5ms TTL=126
Reply from 172.26.0.22: bytes=32 time=5ms TTL=126
Reply from 172.26.0.22: bytes=32 time=9ms TTL=126
Reply from 172.26.0.22: bytes=32 time=10ms TTL=126

Ping statistics for 172.26.0.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 10ms, Average = 7ms
```



COLORADO VLAN 200 VERIFICATION

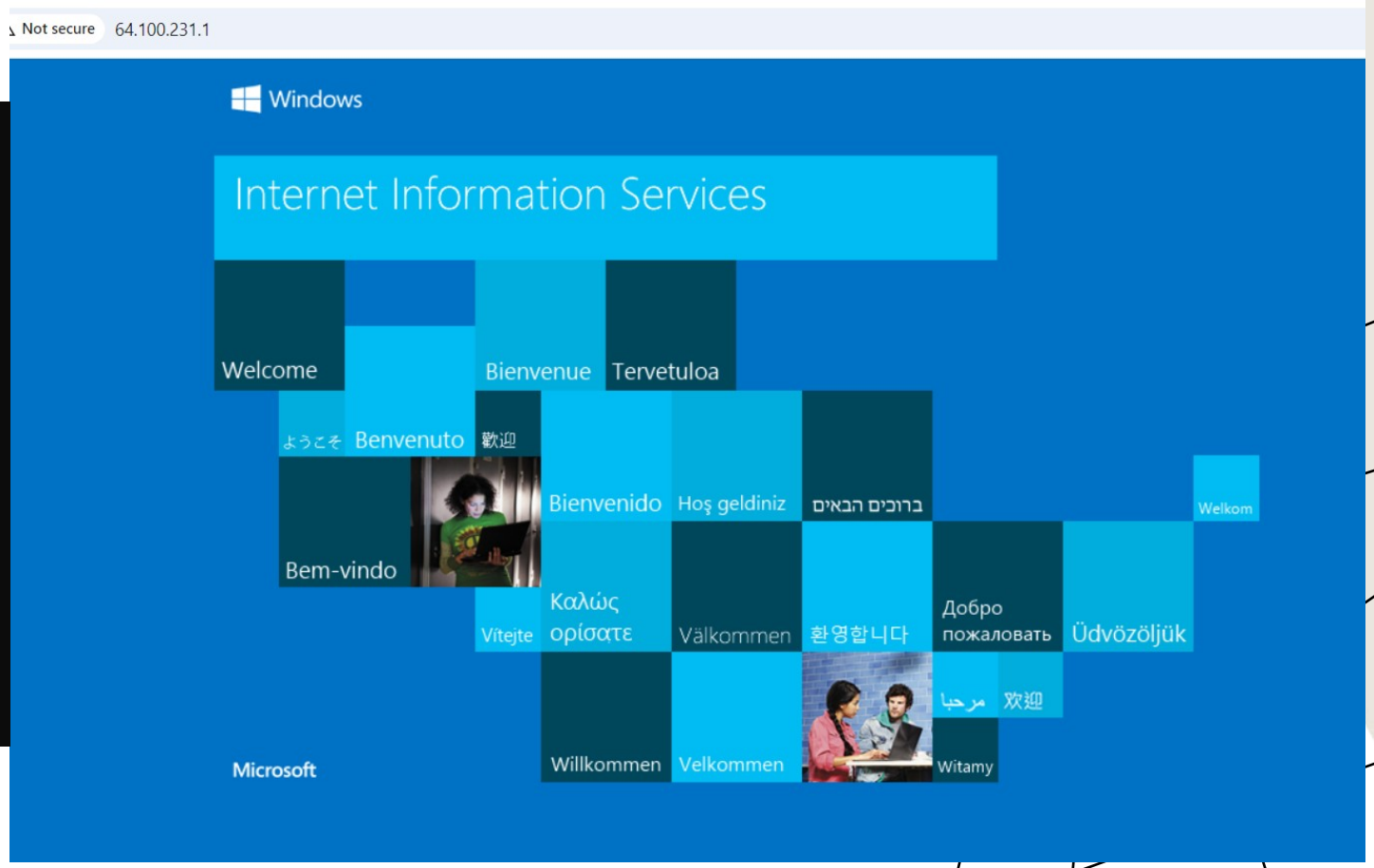
SOURCE	DESTINATION	PROTOCOL	EXPECTED RESULT
VLAN 200	ISP	HTTP	PASS
VLAN 200	LIMA	ICMP	PASS
VLAN 200	SNMP	ICMP	PASS
VLAN 200	SYSLOG	ICMP	PASS
VLAN 200	GERMANY	ICMP	PASS
VLAN 200	VLAN 100	ICMP	PASS
VLAN 200	VLAN 300	ICMP	PASS

COLORADO VLAN 200 VERIFICATION

```
Pinging 64.100.231.1 with 32 bytes of data:
Reply from 64.100.231.1: bytes=32 time=3ms TTL=126
Reply from 64.100.231.1: bytes=32 time=18ms TTL=126
Reply from 64.100.231.1: bytes=32 time=3ms TTL=126
Reply from 64.100.231.1: bytes=32 time=5ms TTL=126

Ping statistics for 64.100.231.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 18ms, Average = 7ms
Pinging 172.26.7.11 with 32 bytes of data:
Reply from 172.26.7.11: bytes=32 time=29ms TTL=125
Reply from 172.26.7.11: bytes=32 time=23ms TTL=125
Reply from 172.26.7.11: bytes=32 time=21ms TTL=125
Reply from 172.26.7.11: bytes=32 time=24ms TTL=125

Ping statistics for 172.26.7.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 21ms, Maximum = 29ms, Average = 24ms
```



COLORADO VLAN 300 VERIFICATION

SOURCE	DESTINATION	PROTOCOL	EXPECTED RESULT
VLAN 300	ISP	HTTP	PASS
VLAN 300	LIMA	ICMP	PASS
VLAN 300	SNMP	ICMP	PASS
VLAN 300	SYSLOG	ICMP	PASS
VLAN 300	GERMANY	ICMP	PASS
VLAN 300	VLAN 100	ICMP	PASS
VLAN 300	VLAN 200	ICMP	PASS

COLORADO VLAN 300 VERIFICATION

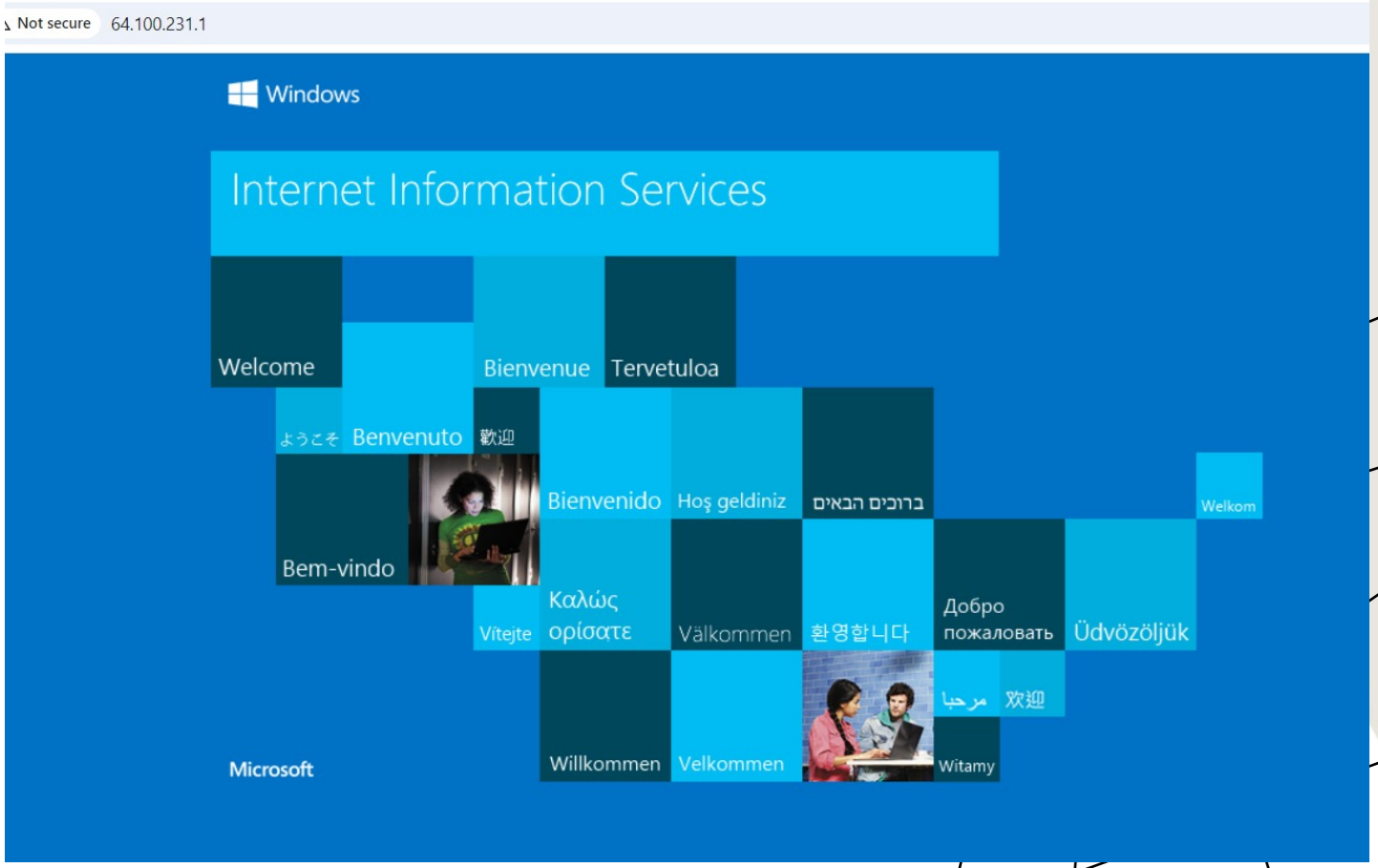
```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : COLO.com
   IPv6 Address. . . . . . . . . . . : 2001:db8:acad:a::38
   Link-local IPv6 Address . . . . . : fe80::4811:8323:da4a:b39c%18
   IPv4 Address. . . . . . . . . . . : 172.26.6.13
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::2%18
                                       172.26.6.1

C:\Users\Cisco Student>ping 172.26.0.22

Pinging 172.26.0.22 with 32 bytes of data:
Reply from 172.26.0.22: bytes=32 time=5ms TTL=126
Reply from 172.26.0.22: bytes=32 time=5ms TTL=126
Reply from 172.26.0.22: bytes=32 time=5ms TTL=126
Reply from 172.26.0.22: bytes=32 time=5ms TTL=126

Ping statistics for 172.26.0.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 5ms, Maximum = 5ms, Average = 5ms
```



LIMA VERIFICATION

SOURCE	DESTINATION	PROTOCOL	EXPECTED RESULT
Host on LIMA	ISP	HTTP	PASS
Host on LIMA	GERMANY	ICMP	PASS
Host on LIMA	SNMP	ICMP	PASS
Host on LIMA	SYSLOG	ICMP	PASS
Host on LIMA	VLAN 100	ICMP	PASS
Host on LIMA	VLAN 200	ICMP	PASS
Host on LIMA	VLAN 300	ICMP	PASS

LIMA VERIFICATION

```
Windows IP Configuration


Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

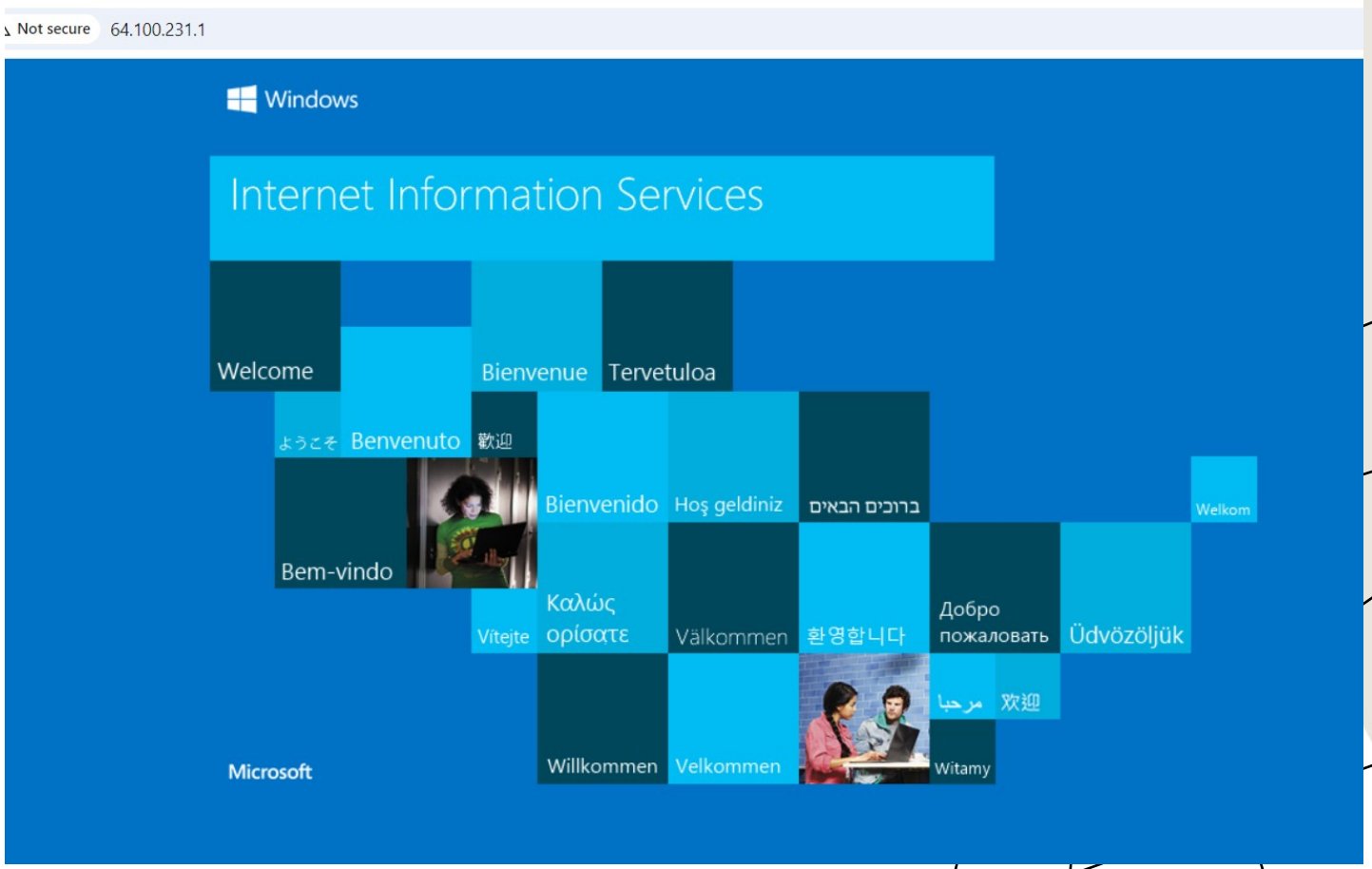
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::2f50:9023:4ed7:3be1%12
   IPv4 Address. . . . . . . . . . . : 172.26.7.11
   Subnet Mask . . . . . . . . . . . : 255.255.255.192
   Default Gateway . . . . . . . . . : 172.26.7.1

C:\Users\Cisco Student>ping 172.26.0.22

Pinging 172.26.0.22 with 32 bytes of data:
Reply from 172.26.0.22: bytes=32 time=11ms TTL=124
Reply from 172.26.0.22: bytes=32 time=14ms TTL=124
Reply from 172.26.0.22: bytes=32 time=61ms TTL=124
Reply from 172.26.0.22: bytes=32 time=11ms TTL=124

Ping statistics for 172.26.0.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 61ms, Average = 24ms
```



BELGUM VERIFICATION

SOURCE	DESTINATION	PROTOCOL	EXPECTED RESULT
SYSLOG	ISP	HTTP	PASS
SYSLOG	GERMANY	ICMP	PASS
SYSLOG	LIMA	ICMP	PASS
SYSLOG	SNMP	ICMP	PASS
SYSLOG	VLAN 100	ICMP	PASS
SYSLOG	VLAN 200	ICMP	PASS
SYSLOG	VLAN 300	ICMP	PASS

SOURCE	DESTINATION	PROTOCOL	EXPECTED RESULT
SNMP	ISP	HTTP	PASS
SNMP	GERMANY	ICMP	PASS
SNMP	LIMA	ICMP	PASS
SNMP	SYSLOG	ICMP	PASS
SNMP	VLAN 100	ICMP	PASS
SNMP	VLAN 200	ICMP	PASS
SNMP	VLAN 300	ICMP	PASS

BELGUM VERIFICATION (SYSLOG)

```
Windows IP Configuration


Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::5880:801a:62cb:8e20%11
   IPv4 Address. . . . . . . . . . . : 172.26.255.238
   Subnet Mask . . . . . . . . . . . : 255.255.255.252
   Default Gateway . . . . . . . . . : 172.26.255.237

Tunnel adapter isatap.{657090E3-4DEC-4BB9-BFE6-8B3454B0E756}:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

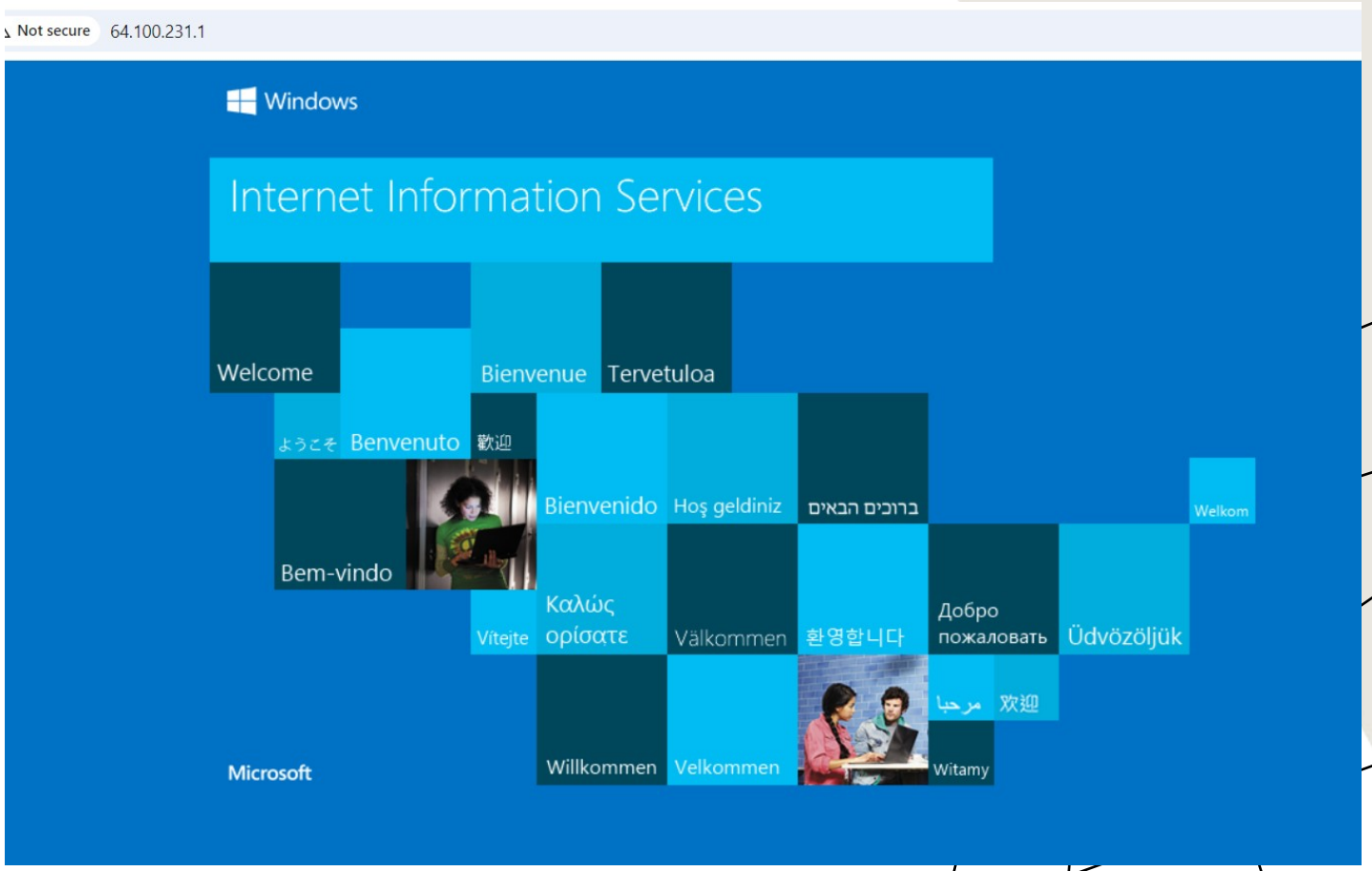
Tunnel adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

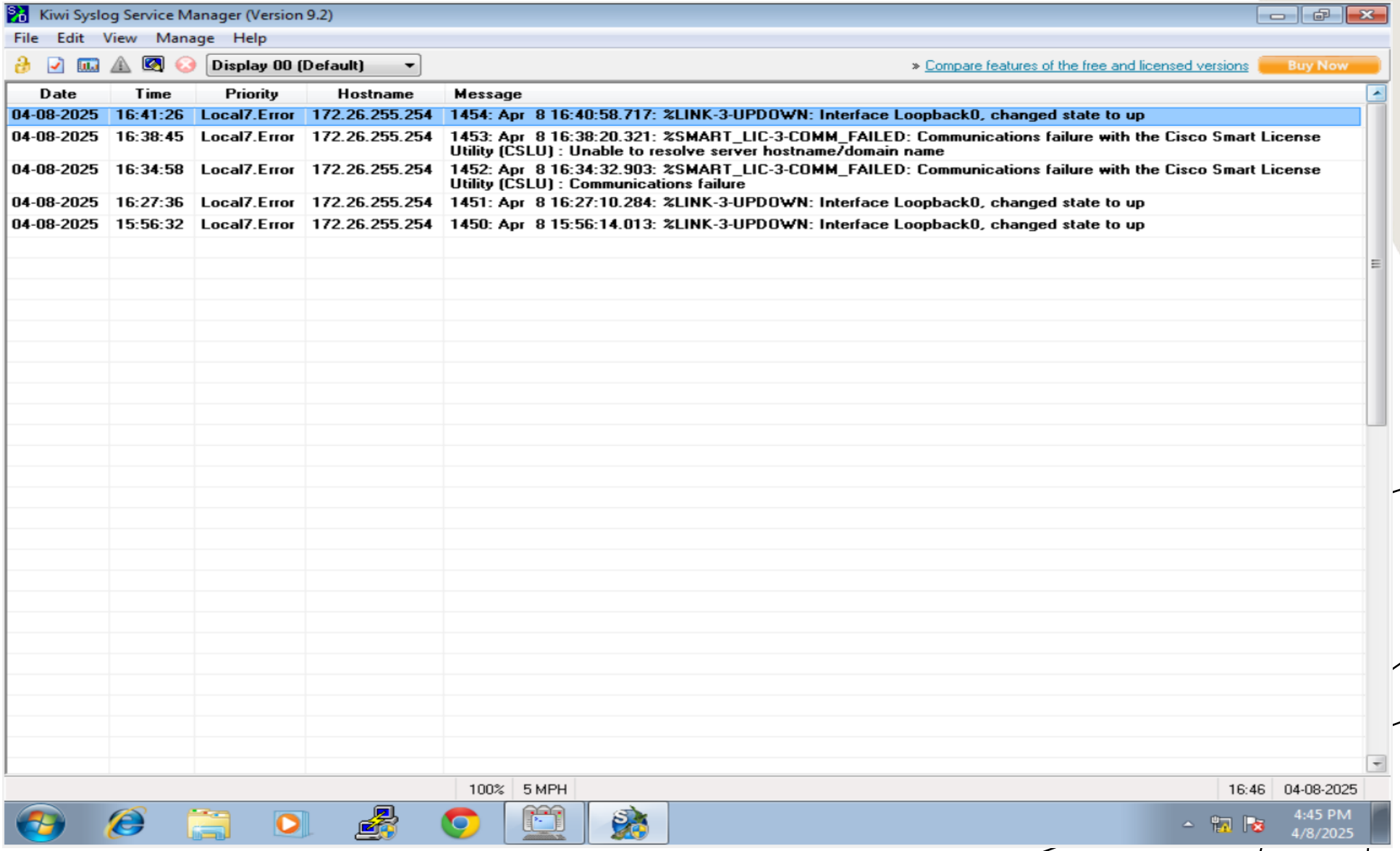
C:\Users\Cisco Lab>ping 172.26.0.22

Pinging 172.26.0.22 with 32 bytes of data:
Reply from 172.26.0.22: bytes=32 time=6ms TTL=125
Reply from 172.26.0.22: bytes=32 time=8ms TTL=125
Reply from 172.26.0.22: bytes=32 time=7ms TTL=125
Reply from 172.26.0.22: bytes=32 time=10ms TTL=125

Ping statistics for 172.26.0.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 10ms, Average = 7ms
```



SYSLOG VERIFICAITON



BELGUM VERIFICATION (SNMP)

```
Windows IP Configuration


Ethernet adapter Local Area Connection:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2001:acad:db8:2:4::5
   Link-local IPv6 Address . . . . . : fe80::7141:684b:88e2:10a1%11
   IPv4 Address. . . . . . . . . . . : 172.26.255.242
   Subnet Mask . . . . . . . . . . . : 255.255.255.252
   Default Gateway . . . . . . . . . : 2001:acad:db8:2:4::1
                                       172.26.255.241

Tunnel adapter Local Area Connection* 9:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

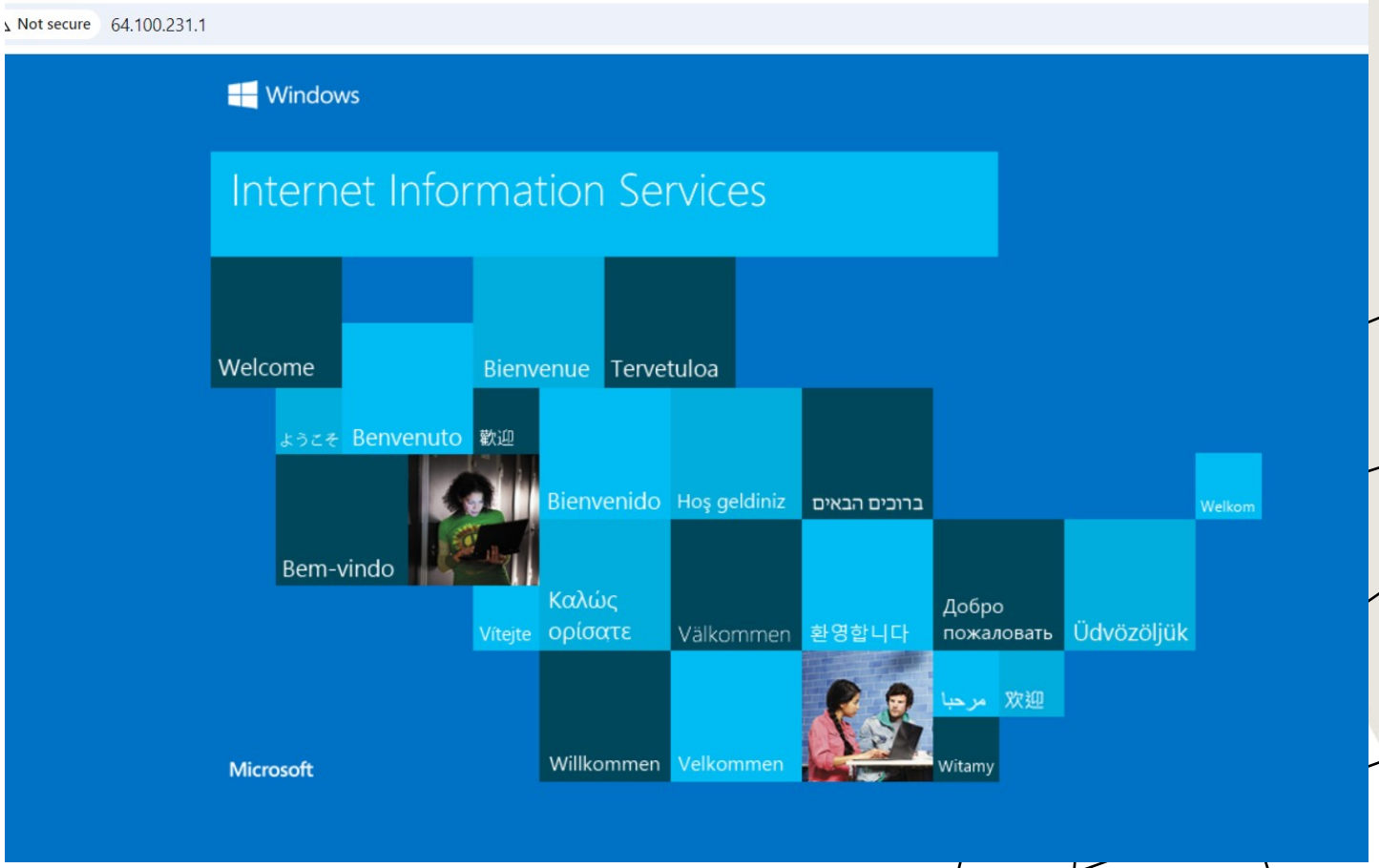
Tunnel adapter isatap.{657090E3-4DEC-4BB9-BFE6-8B3454B0E756}:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

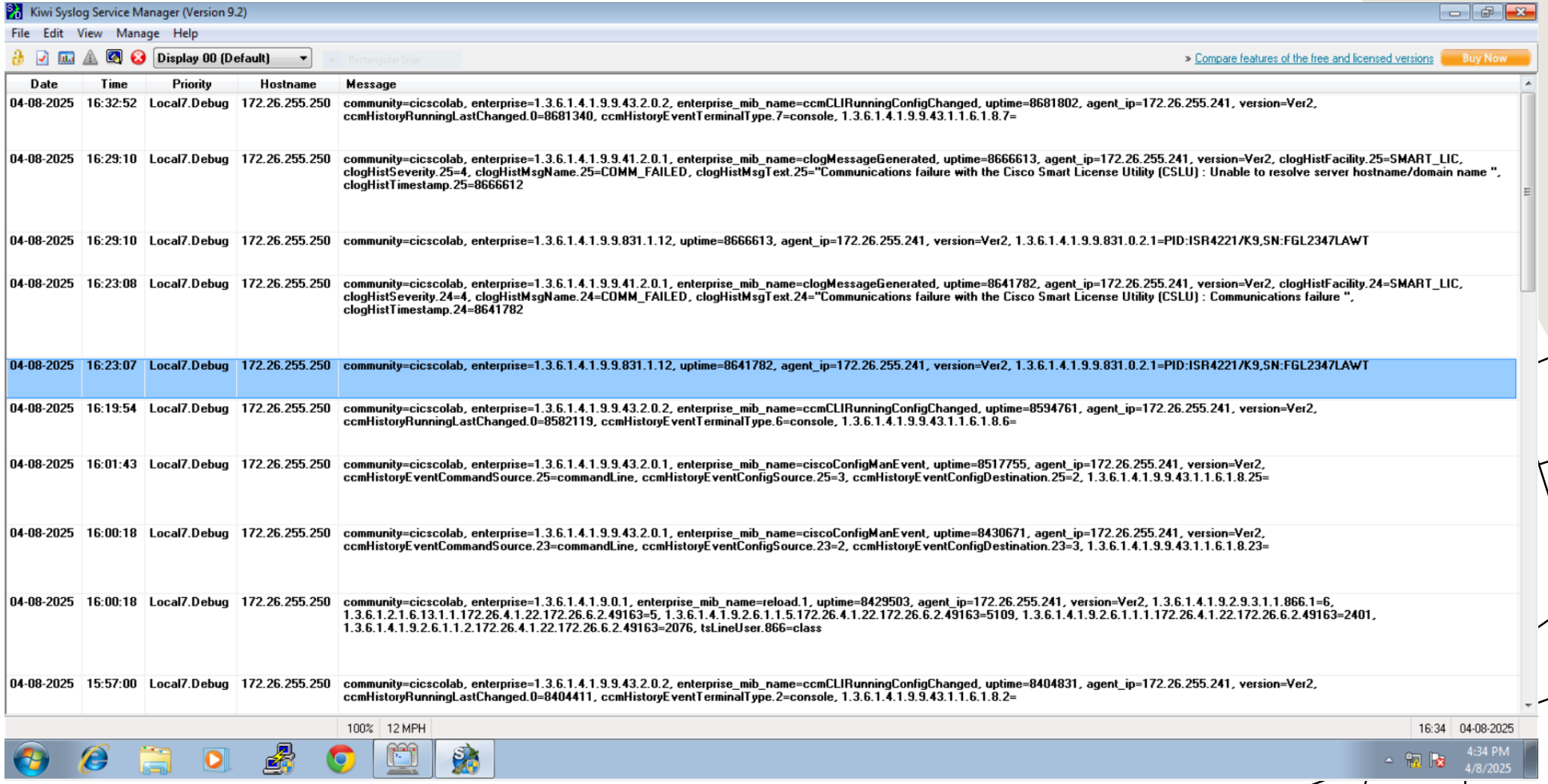
C:\Users\Cisco Lab>ping 172.26.0.22

Pinging 172.26.0.22 with 32 bytes of data:
Reply from 172.26.0.22: bytes=32 time=6ms TTL=125
Reply from 172.26.0.22: bytes=32 time=6ms TTL=125
Reply from 172.26.0.22: bytes=32 time=7ms TTL=125
Reply from 172.26.0.22: bytes=32 time=8ms TTL=125

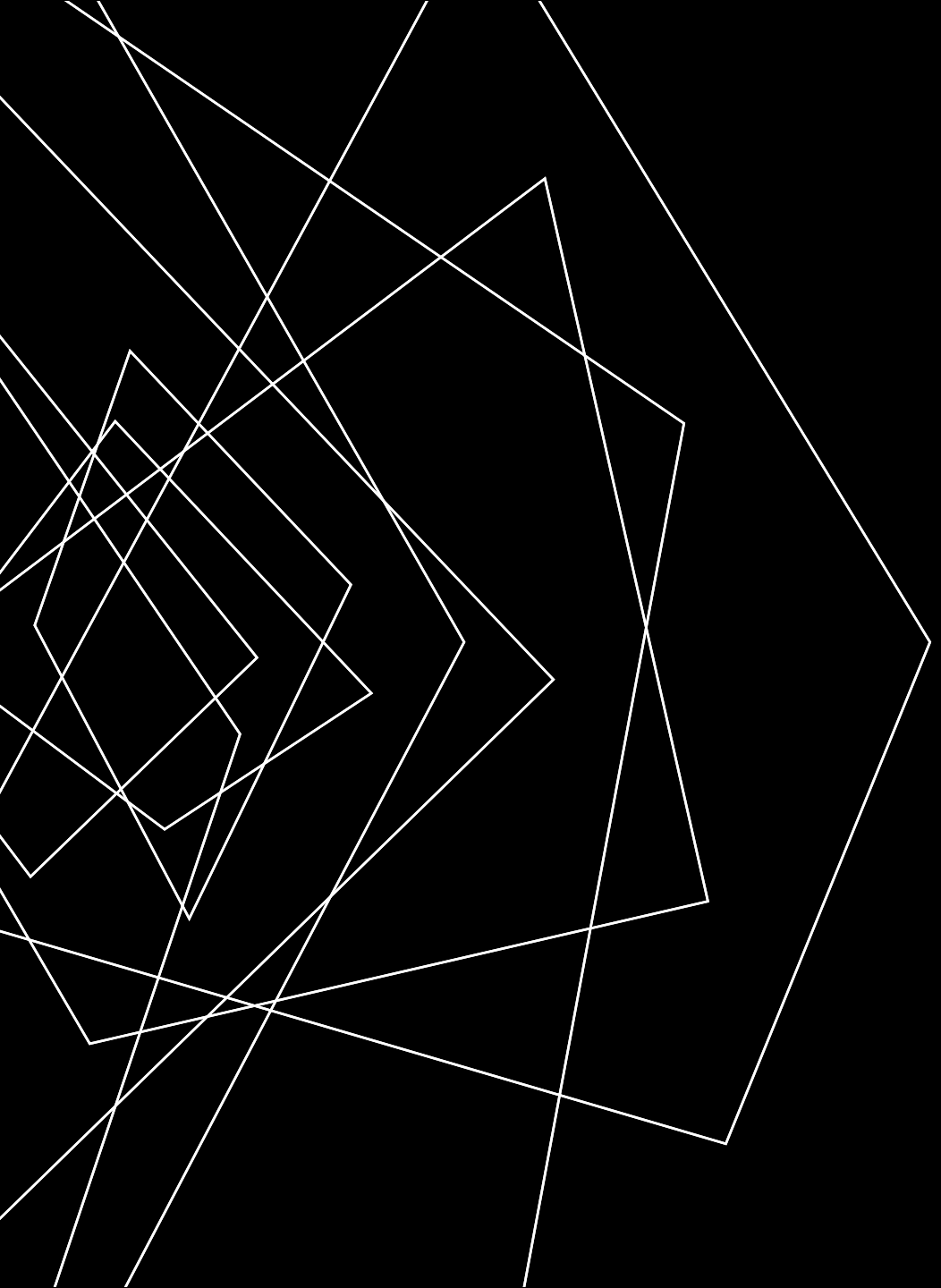
Ping statistics for 172.26.0.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 8ms, Average = 6ms
```



SNMP VERIFICAITON







THANK YOU

Darren Huynh